

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Results-driven **Quality Assurance / Test Automation Engineer** with **6+ years of experience** designing test strategies, automation frameworks, and validation solutions for backend services, APIs, distributed systems, enterprise applications, and customer-facing digital platforms.
- Strong programming and automation experience using **Python, Java, Kotlin, JavaScript, SQL, REST APIs, and workflow-based test harnesses** to exercise complex backend processes and distributed application behavior.
- Hands-on expertise designing and executing automated tests for **APIs, services, data models, backend workflows, integration layers, and client-to-service communication**, with strong understanding of API documentation and specifications.
- Experienced building automated workflow simulations for **multi-agent coordination, resource allocation, planning, scheduling, optimization, and distributed service interactions** across scalable backend systems.
- Strong production-debugging experience analyzing **backend logs, application traces, API traffic, database records, and telemetry** to triage defects, reproduce failures, perform root-cause analysis, and create actionable bug reports.
- Experienced validating **scalability, observability, fault tolerance, reliability, concurrency, data integrity, and failure-recovery behavior** across cloud-native and containerized environments using Kubernetes and Docker.
- Proven ability to independently lead release testing, defect triage, root-cause analysis, process improvement, and cross-functional quality initiatives while collaborating with developers, Product, DevOps, and business stakeholders.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Scripting:** Python, Java, Kotlin, JavaScript, TypeScript, C#, SQL, Bash, Workflow Automation Scripts
- **Backend & API Testing:** REST APIs, API Specifications, Postman, REST Assured, Swagger/OpenAPI, JSON, Microservices, Service Integration, API Contract Validation
- **Automation & Workflow Validation:** Playwright, Selenium WebDriver, pytest, TestNG, JUnit, Cucumber BDD, Multi-Agent Workflow Simulation, Resource Allocation, Planning, Optimization
- **Distributed Systems Quality:** Backend Services, Distributed Applications, Scalability Testing, Fault-Tolerance Testing, Reliability Testing, Concurrency, Service Communication, End-to-End Validation
- **Observability & Debugging:** Backend Log Analysis, Application Traces, Telemetry, Splunk, Root Cause Analysis, Defect Triage, Failure Reproduction, Bug Reporting
- **Cloud & Containers:** Kubernetes, Docker, AWS, Azure, Cloud-Native Applications, Containerized Services, CI/CD, Jenkins, GitLab, GitHub
- **Data & Delivery:** Large Data Sets, SQL, PostgreSQL, MySQL, Oracle, Data Validation, Release Testing, Process Improvement, Agile Scrum, Jira, Cross-Functional Collaboration

EXPERIENCE

Test Automation Engineer **PCMI LLC (Full Time)**

Oct 2024 – Present
San Francisco, CA, USA

- Designed and implemented comprehensive test plans and automated validation strategies for distributed backend services, APIs, data models, and enterprise workflows, covering functional, integration, system, regression, and end-to-end testing.
- Developed **Python-based automation scripts and workflow simulators** to exercise complex service interactions, concurrent transactions, resource allocation, scheduling, planning, and multi-agent-style coordination across distributed backend systems.
- Built API and service-level automation using Python, REST Assured, Postman, Swagger/OpenAPI, and JSON, validating request/response contracts, business rules, authentication, error handling, service dependencies, and client-to-backend communication.
- Validated scalable and fault-tolerant application behavior by simulating service failures, retries, timeouts, concurrent workloads, degraded dependencies, recovery scenarios, and high-volume interactions across cloud-native environments.
- Analyzed backend logs, service traces, API traffic, SQL records, and application telemetry to independently triage complex defects, reproduce intermittent failures, perform root-cause analysis, and provide detailed evidence in defect reports.
- Tested containerized services deployed through **Kubernetes and Docker**, validating service discovery, configuration, deployment health, inter-service communication, scaling behavior, resiliency, observability, and production readiness.
- Executed release-testing and quality-gate activities across CI/CD pipelines, monitoring regression results, defect trends, backend stability, and operational quality signals while identifying risks prior to production deployment.
- Partnered closely with developers, architects, Product Owners, DevOps, and business stakeholders to improve application testability, observability, system reliability, and fault tolerance while driving defect triage, RCA, and continuous process improvement.

Environment: Python, Java, Kotlin, REST APIs, Postman, REST Assured, Swagger/OpenAPI, JSON, Backend Services, Distributed Systems, Multi-Agent Workflow Simulation, Resource Allocation, Planning, Optimization, SQL, PostgreSQL, Backend Logs, Telemetry, Splunk, Kubernetes, Docker, AWS, CI/CD, Jenkins, Jira, RCA, Release Testing, Agile Scrum

SDET

Dec 2023 – Sept 2024
Remote, USA

Client: CVS

- Designed and executed test strategies for large-scale healthcare backend services, APIs, distributed applications, and customer-facing workflows, validating service functionality, reliability, data integrity, and end-to-end system behavior.
- Developed Python automation and workflow-simulation utilities to validate complex distributed business processes involving scheduling, resource assignment, service coordination, asynchronous processing, and optimization across backend platforms.
- Automated REST API and service validation using Postman, REST Assured, Swagger/OpenAPI, JSON, and Python, verifying API specifications, payload schemas, service contracts, authentication, error handling, and downstream integrations.
- Validated communication between client applications and backend services by exercising mobile/web-to-service workflows, service dependencies, transaction state changes, retry logic, timeout handling, and degraded-service scenarios.
- Performed deep backend triage using **Splunk logs, application traces, API responses, SQL queries, and telemetry**, reproducing complex production-like issues and documenting actionable root-cause findings for engineering teams.
- Tested scalability, fault tolerance, and observability of cloud-native services running in Kubernetes and Docker environments, including load behavior, container health, service recovery, deployment configuration, and distributed dependencies.
- Processed and validated large healthcare datasets using SQL and automated data checks, ensuring consistency between service requests, backend processing, database state, API responses, and downstream workflows.
- Led regression and release validation while collaborating with developers, platform engineers, Product, and DevOps teams to resolve high-priority defects, improve testability, strengthen observability, and enhance overall system reliability.

Environment: Python, Java, Kotlin, REST APIs, Postman, REST Assured, Swagger/OpenAPI, JSON, Distributed Systems, Backend Services, Workflow Simulation, Large Data Sets, SQL, MySQL, PostgreSQL, Splunk, Log Analysis, Telemetry, Kubernetes, Docker, AWS, Jenkins, CI/CD, RCA, Release Testing, Jira, Agile Scrum

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Designed and executed automation and integration testing for enterprise backend applications, APIs, distributed services, and data-driven workflows using Java, Python, Selenium WebDriver, REST services, and SQL.
- Developed reusable Python and Java automation utilities to exercise complex application workflows, backend processing, asynchronous operations, service integrations, and end-to-end business scenarios.
- Performed API validation using Postman, SoapUI, REST services, JSON/XML payloads, and API documentation, verifying request/response contracts, error handling, authentication, and integration behavior.
- Validated distributed application behavior across frontend, service, API, and database layers, testing transaction consistency, service dependencies, failure handling, recovery paths, and integration reliability.
- Analyzed application and backend logs, API responses, SQL records, and execution results to reproduce defects, isolate root causes, identify higher-risk components, and collaborate with developers through resolution.
- Performed data validation against large relational datasets using SQL, verifying backend transactions, business rules, application state, service outputs, and data consistency throughout integrated workflows.
- Integrated automated suites with Jenkins, Git, and CI/CD pipelines to support continuous regression execution, release validation, rapid feedback, and reliable software deployment across development environments.
- Collaborated with Product Managers, Software Engineers, QA teams, and technical stakeholders to improve test coverage, application testability, defect prevention, release quality, and continuous testing processes.

Environment: Python, Java, REST APIs, Postman, SoapUI, JSON, XML, Selenium WebDriver, TestNG, Distributed Systems, Backend Testing, SQL, Oracle, MySQL, Log Analysis, Jenkins, Git, CI/CD, Regression Testing, Integration Testing, RCA, Jira, Agile Scrum

Associate Engineer QA

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed and executed automated test suites for enterprise web, mobile, API, and backend systems using Java, Python, Selenium WebDriver, Appium, Cucumber BDD, TestNG, and REST-based services.
- Created automation scripts to exercise end-to-end workflows spanning client applications, backend APIs, databases, and integrated services, validating functional behavior and system reliability across multiple application layers.
- Performed API testing using REST services and Postman, validating API specifications, request/response payloads, authentication, error scenarios, business rules, and downstream service behavior.
- Executed system, integration, regression, functional, and end-to-end testing across distributed application components while validating communication between mobile clients, backend services, and enterprise systems.
- Analyzed application logs, backend responses, SQL data, and execution evidence to triage defects, reproduce complex issues, support root-cause analysis, and provide detailed defect documentation.
- Performed SQL-based data validation to verify application transactions, backend state changes, business logic, and data integrity across integrated workflows and large enterprise datasets.
- Integrated test automation with Jenkins, Maven, Git, and CI/CD pipelines to support repeatable validation, continuous regression testing, build verification, and production-release readiness.
- Collaborated with developers, Business Analysts, QA Leads, and Product stakeholders to improve testing processes, increase automation coverage, investigate integration issues, and deliver reliable releases within Agile sprint cycles.

Environment: Java, Python, Selenium WebDriver, Appium, Cucumber BDD, TestNG, REST APIs, Postman, JSON, SQL, Backend Testing, Distributed Applications, System Integration Testing, Log Analysis, Jenkins, Maven, Git, CI/CD, RCA, Regression Testing, Jira, Agile Scrum

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.